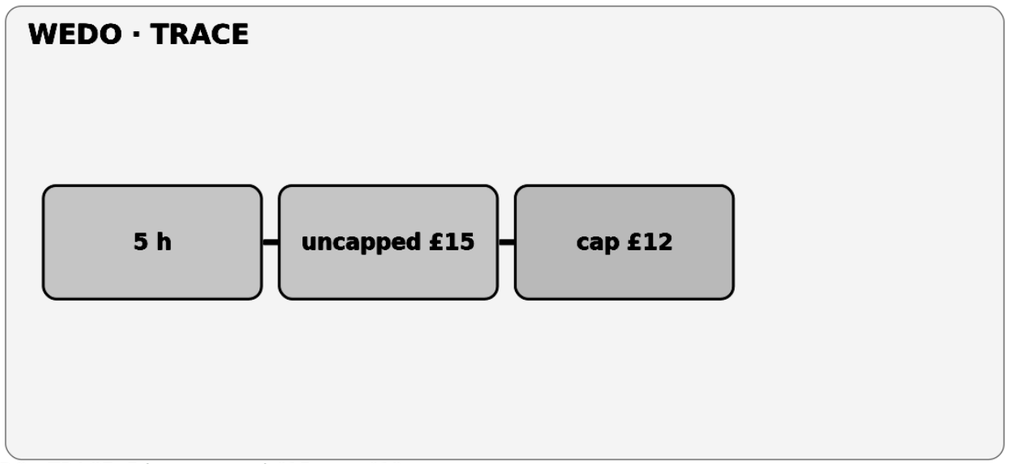
We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.

Cap is new relevant rule. Example 5 hours costs £12, not £15.



WEDO · TRACE: 5 h; uncapped £15; cap £12

DO NOW — RECALL

Lesson 060 · Refining an Abstraction and Testing Assum

Name: _____ Class: _____ Date: _____

1. What does abstraction remove?

2. What is a model assumption?

3. What is a boundary test?

4. Two rising-edge D-type flip-flops A and B share a clock and start with QA=0 and QB=0. DA=1 throughout, and DB is connected to QA. Explain why QB is still 0 immediately after the first rising edge. Assume ideal simultaneous sampling.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Irrelevant details.

2. Model answer

Simplifying rule treated as true for a task.

3. Model answer

An input at/around an important threshold.

4. Model answer

Both sample their inputs at the same edge. B samples the old QA=0, while A samples 1. A's new output reaches B after that sampling event.

By the end, you can…

• Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

• Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

• Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Vocabulary — know these terms

refinement — Improving model for new requirements.

boundary — Important threshold or edge.

validity — Whether model works for its intended use.

assumption — Rule taken as given.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Explain
Cap is new relevant rule. Example 5 hours costs £12, not £15.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A model that worked yesterday never needs changing when requirements change.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

NEW LEARNING

Read, annotate and explain

Understand

Compare the simplification against new requirements. If the model omits a relevant factor it can produce plausible but wrong outputs.

Model

Refine the abstraction by adding only the newly necessary variable, rather than reconstructing every real-world detail.

Evaluate

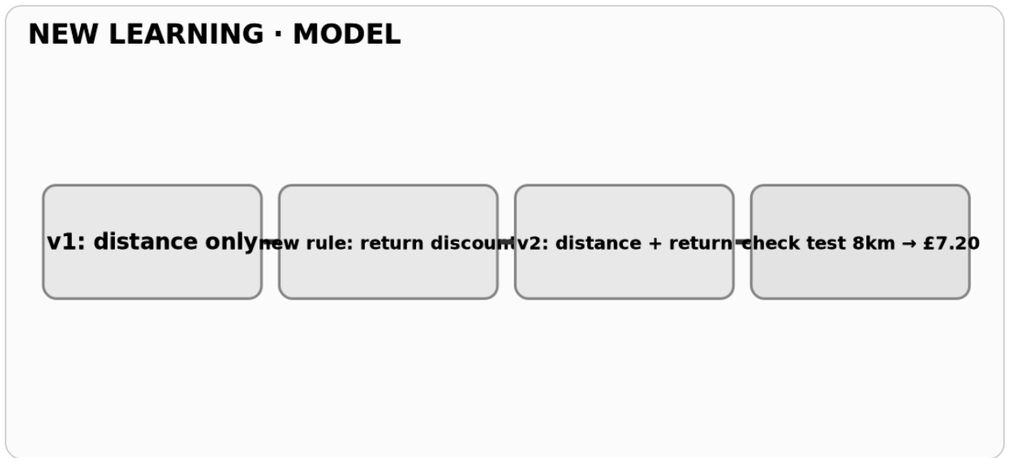
Test normal, boundary and exceptional cases: 5 km versus 6 km, return and single, and invalid negative distance.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



NEW LEARNING · MODEL: v1: distance only; new rule: return discount; v2: distance + return; check test 8km → £7.20

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

A school model predicts homework concerns from just missed work counts; discuss which relevant conditions to add and how to test impact. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Ticket v1 charges £2 at ≤5km, £4 otherwise. Give charge at 5km and 6km and why these are useful tests. [6]

Question 2

Return journeys cost 1.8 times single. For 8km journey calculate return fare and show variables. [6]

I DO — WORKED EXAMPLE

Follow the worked method

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

Explain

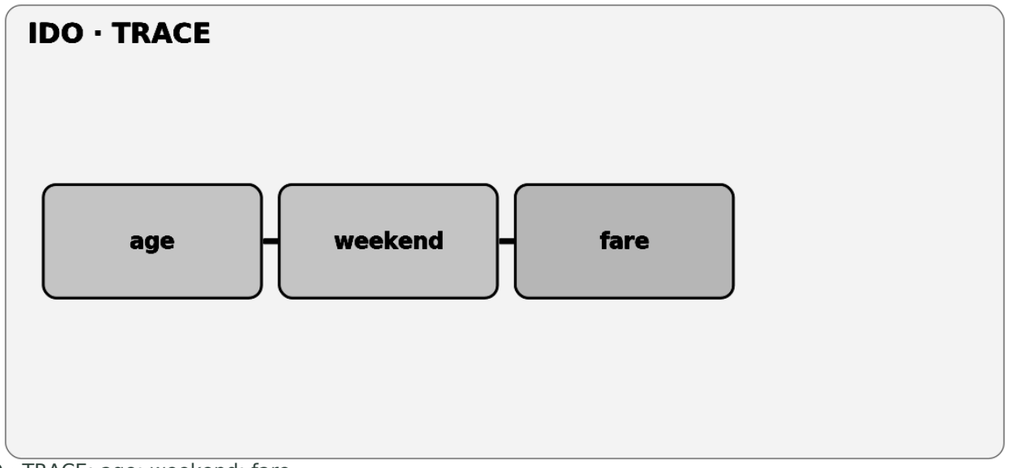
Weekend becomes a relevant distinction because it can change the output.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A simplified cinema ticket model uses only age but now weekend prices differ. What change is needed?

Weekend becomes a relevant distinction because it can change the output.



IDO · TRACE: age; weekend; fare

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A parking price model charges £3/hour but introduces £12 maximum. Improve algorithm.
Source code / input:
fee=MIN(3×hours,12), with validation that hours≥0.